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Pulse Test: Underutilized Technology for Reservoir Management and Connectivity Understanding. Untapping the Technology with Signal Processing Techniques

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Abstract

Interwell connectivity evaluation is a fundamental component of reservoir management in both mature and undeveloped fields. This is particularly critical in geologically complex settings characterized by faulting, stratigraphic discontinuities, and potential compartmentalization. Pulse testing, which involves generating pressure pulses in a generator well and observing pressure responses in other wells (observation wells), presents a cost-effective method for assessing connectivity because no producers are shut-in (C.R. Johnson; R.A. Greenkorn). However, its practical implementation is often constrained by signal detection challenges—especially in low-permeability formations, reservoirs with high gas saturation, or when wells are widely spaced. In such cases, pressure signals become weak and easily obscured by operational noise or unrelated reservoir dynamics, making conventional interpretation approaches such as transient pressure analysis or numerical simulation less effective.

This paper presents a dedicated pulse test workflow developed to overcome these limitations. The workflow integrates advanced signal processing techniques with analytical modelling to ensure both detectability and interpretability of pulse responses in complex and challenging environments. A key part of the workflow, critical for pulse test design and interpretation, is based on signal decomposition, statistical correlation techniques, and analytical models. In the interpretation stage, the method decomposes measured pressure data into signal components attributable to the pulsing well and background noise. By applying diffusion-based mathematical models to isolate and match the response, the workflow quantifies interwell transmissibility and storativity, while also estimating associated uncertainty based on the signal-to-noise ratio. The automated interpretation workflow consists of three main stages:

1. Preprocessing: Removes background trends and noise using detrending, highlighting periodic features in gauge data,
2. Signal Decomposition: Splits pressure data into two parts—one attributed to the pulsing well, the other to unrelated events – allowing a cleaner interpretation
3. Parameters Estimation: Calculates interwell transmissibility and storativity, with uncertainty linked to the signal-to-noise ratio and connectivity confidence.

The methodology was applied in Field A, a clastic onshore oil reservoir with compartmentalization and an expanding gas cap. The pulse test campaign included three 240-hour cycles involving one injector and three producer-observation wells. The proposed signal decomposition approach enabled the identification of pressure responses with up to 86% correlation to the pulsing sequence, overcoming interference from operational noise and variable reservoir conditions. The derived time lags and amplitudes were used to compute interwell transmissibility, providing a quantified measure of reservoir connectivity across fault blocks. These findings supported decision making related to optimization of the injection strategy and justified new well placement decisions for a planned infill injector.

This work demonstrates that automated signal processing significantly enhances the reliability and applicability of pulse test interpretation in reservoirs where the application of traditional methods is challenging. The workflow enables:

- Interpretation in high gas saturation and compartmentalized reservoirs previously considered unsuitable for pulse testing.
- Quantitative evaluation of signal presence and strength, offering confidence metrics for interwell connectivity.
- Operational efficiency, with no need for complex forward modelling or heavy manual curve matching.
- Scalability, supporting multi-well, multi-cycle pulse test campaigns.

Ultimately, this paper provides a methodology and general workflow that extends the utility of pulse testing into more challenging reservoir environments by leveraging signal decomposition and analytical modelling.

Objectives

The paper aims to demonstrate a practical and versatile pulse-testing methodology that overcomes the long-standing signal detection limitations that have restricted the technique to high-permeability, closely spaced well pairs. By coupling automated signal decomposition with diffusion-based interpretation, the approach quantifies interwell transmissibility and storativity-even in reservoirs marked by complex faulting, wide well spacing, low or moderate permeability, and high gas saturation.

Although pulse testing is often associated with water-injection projects, the technique is equally applicable in gas reservoirs. A dedicated injector is not a prerequisite: any producing well can be cycled to provide the pressure disturbance, while neighbouring wells-fitted with high-resolution gauges-monitor the response. This capability is rarely exploited, which partly explains why pulse testing remains underutilised in gas-dominated fields and in mature oil assets without injection system or in which injectors have been shut in. Thus either producers or injectors can act as the pulse source broadens the reach of the technology and opens new opportunities for connectivity surveillance across a wider range of reservoir types.

The paper focuses on establishing a repeatable workflow that removes background trends, isolates the pulse-coherent signal, and directly estimates transmissibility/storativity with quantified uncertainty. The study positions pulse testing as a practical surveillance tool for a much broader spectrum of reservoirs than previously considered feasible.

Methodology

Today there are several industry applied solutions of signal processing techniques (Béla Tóth, Mihály Megyery). The paper introduces automated solution for interpreting pulse tests by employing a pressure decomposition approach as an alternative to the traditional method. The signal-based decomposition method

allows for the disassembly of the cumulative pressure response at each well into individual components. This aids in identifying signals originating from the pulsing well.

Each well pair (pulsing and observation) is modeled using a 1D line-source solution for pressure diffusion. Reservoir properties including permeability (k), porosity (ϕ), effective thickness (h), fluid viscosity (μ), total compressibility (ct), and interwell distance (r) are input as parameters for each zone. From these, the dimensionless coefficients a and b for the exponential integral solution are derived. Two physical parameters characterize the interwell response: transmissibility (T), which quantifies the ease of fluid communication between wells, and storativity (S), which relates to the ability of the reservoir to store fluid and respond to pressure changes.

$$a = \frac{70.6B\mu}{kh}, b = \frac{948\phi\mu c_t r^2}{k}, T = \frac{kh}{\mu}, S = \phi h c_t$$

A single on/off pulse is described by the classical line-source solution:

$$\Delta p(t) = -aQEi\left[-\frac{b}{t-t_0}\right]$$

where $\Delta p(t)$ is the pressure change at the observation well, t_0 is the cycle start time, Q is rate (positive for production, negative for injection) and Ei is the exponential integral.

This analytic solution allows fast generation of pressure responses for multiple on/off cycles of pulsing activity, using variable cycle durations and injection/production rates.

By leveraging diffusion models within the signal's solution space, the technique searches for the optimal solution with the lowest correlation function for each well. This process helps in detecting signal presence and estimating transmissibility and storativity parameters, based on the transient response to the signal generated at the injector well.

The technique workflow includes pressure data preprocessing (detrending, etc.), pressure data decomposition, and evaluation of interwell reservoir parameters between pulsing and observation wells, including transmissibility and storativity in the range of uncertainty based on signal/noise ratio and connectivity between pulsing and observation wells.

The first step is preprocessing the pressure data to accentuate its periodic features through noise reduction and detrending. This initial step aims to eliminate the influence of steady-state pressure responses and diminish pressure disturbances' impact during testing, guided by an analysis of gauge history.

The second step is pressure decomposition for dissecting the pressure response elements corresponding to signals generated by the pulsing well. This method employs a mathematical algorithm to split the pressure variations observed in the receiving well into two distinct components. The first component is aligned with the pulsing well's contribution, serving as the baseline response, while the second component accounts for variations related to other events. Through this decomposition, the algorithm aims to minimize the correlation function between pressure variation in observation well and the estimated responses to the pulse. This phase effectively distinguishes between pressure changes caused by the pulsing well's signal and other influences, allowing for automated matching across multiple wells.

The final stage is calculating the interwell reservoir parameters between the pulsing and observation wells. This includes determining transmissibility, storativity, and assessing uncertainty ranges, based on the signal-to-noise ratio and the connectivity level between the wells.

General Workflow for Pulse Test Operations

A pulse test workflow to safeguard data quality and maximise diagnostic value:

1. **Well Selection and Readiness.** Candidate pulsing and observation wells are screened for integrity, accessibility, and gauge deployability. Any behind-casing leakage risk is mitigated, and slickline resources, down-hole gauge strings, batteries, and hangers are mobilised.

2. **Pulse-Signal Design.** Using estimated reservoir properties and measured background noise, engineers evaluate the pressure pulse so that its amplitude exceeds the gauge noise floor. The number of cycles, on/off durations, and rate are evaluated analytically-and, where necessary, refined with a 2-D sector model-to ensure each pulse is distinguishable at the observation wells.
3. **Programme Execution.** After a short pre-test stabilisation, the planned sequence of rate steps (or choke changes for a producer) is applied. Continuous down-hole pressure is logged in both the pulsing well and the receivers. Operational discipline is critical: any unscheduled rate change introduces noise that can mask the signal.
4. **Data Processing and Interpretation.** Raw pressures are filtered and detrended to remove tidal, barometric, and depletion effects. Automated signal decomposition isolates the pulse signal component, from which interwell transmissibility and storativity with uncertainty are calculated.

Well Selection, Integrity Assurance, and Logistics Readiness

The first phase of any pulse testing campaign is a comprehensive readiness review of both the pulsing and observation wells. This preparation is crucial: weak or noisy signals cannot be interpreted later by sophisticated processing if the basic data acquisition operations are compromised.

The process begins with integrity and injectivity or deliverability verification. Casing and completion integrity are reviewed by examining recent caliper, MIT/pressure tests, and cement bond logs to confirm that no corrosion or mechanical damage could jeopardize confined flow. To rule out behind casing flow or cross flows, temperature and noise logs are assessed. If ambiguity remains, targeted slickline plugs or straddle packers are proposed as part of the intervention plan. Additionally, injectivity or the capability for controlled rate changes is confirmed via step-rate or mini falloff tests, ensuring the pulsing well can deliver the planned flow rates.

Next, slickline and pressure gauge readiness is ensured. Slickline winches, lubricators, and pressure control equipment are pressure-tested and certified for the anticipated shut-in pressure. Each well is equipped with dual quartz gauges to ensure continuous data recording even in the event of battery or sensor failure. The selected gauges must meet the required specifications for accuracy, drift, and temperature rating, and battery sizing is validated to last at least 20 % longer than the total pulsation period.

Instrument hanger assembly follows, during which a complete hanger string-including PXN/PXX plug, crossover sub, shock absorber, and wireline head-is bench assembled and function-tested. On-site expertise is also critical. A dedicated field engineer accompanies the slickline crew to oversee the deployment of equipment, verify gauge functionality, synchronize system clocks, and perform a zero-load drift test to validate measurement reliability.

Finally, clear communication and scheduling are essential to success. Daily coordination meetings between the wells team, production operations personnel, and the slickline contractor align the pulse test schedule with other ongoing field activities. A lock-out/tag-out procedure is also issued to production operators to prevent any unintended choke manipulations or rate changes during the test period.

Pulse Test Design

Objective is to evaluate a pulse sequence that produces a detectable, diagnostic pressure response at each observation well while remaining operationally practical. The design workflow integrates noise analysis, analytic forecasting, and, where required, sector-model validation to fine-tune cycle geometry and rate magnitudes.

The process begins with a noise and detectability assessment. At least one week of pre-test gauge data is filtered to characterize background fluctuations stemming from barometric pressure changes, tidal effects, and routine well operations. The root-mean-square (RMS) noise level at the dominant tidal frequency is established as a benchmark. For the pulse signal to be reliably identifiable during post-processing, its

amplitude must exceed at least twice the RMS noise level, a threshold that typically ensures a correlation coefficient of 0.7 or greater.

Next, reservoir uncertainty is addressed by preparing a range of values for key parameters such as permeability, porosity, total compressibility, and gas saturation. Analytic forward modeling is then used to forecast signal arrival times and pressure amplitudes for each combination of these inputs. This helps verify that even under pessimistic reservoir conditions, the pulse signal remains above the detectability threshold.

The design of the pulse sequence—including the number of pulses, active duration, and shut-in duration—is calculated to enhance signal clarity and minimize interference. To avoid aliasing during signal decomposition, no two pulse cycles may be designed with identical lag times (M. Kamal, W.E. Brigham). A staggered sequence—such as 8 hours on followed by 12 hours off, 10 hours on followed by 16 hours off, and 12 hours on followed by 18 hours off—is preferred to prevent overlap between pulses and improve the ability to identify connectivity parameters. A Monte Carlo simulation with over 1,000 realizations is performed to quantify the distribution of arrival times and amplitudes. The pulse design is iterated until at least 90% of simulated scenarios exceed the minimum detectability threshold.

Finally, when potential geological features such as sealing faults, stratigraphic barriers, or thief zones are suspected, a 2D or 3D sector model is designed to simulate the planned pulse design execution. This allows for validation of pulse behavior under realistic reservoir geometries and provides confidence that the test design remains viable in the presence of complex subsurface conditions.

Pulse-Test Execution and Field Surveillance

The pulse test is executed under tightly controlled field conditions to ensure clean and interpretable data while maximizing the signal-to-noise ratio. The process begins with baseline pressure monitoring. Pressure gauges are deployed and record continuously for at least two weeks prior to the initiation of pulsing. This baseline monitoring period captures natural variations such as tidal loading, barometric fluctuations, and any low-frequency production-related pressure trends. These data are used to establish a quantitative noise profile that serves as a reference against which the amplitude of the pressure response from the pulse will later be evaluated.

Following the baseline monitoring, the pulsing cycles in the selected pulsing well are performed according to the pre-designed signal programme. Synchronization of clocks and key events between the pulsing and observation wells is ensured to allow precise interpretation of pulse arrival and amplitude. During the active testing period, strict isolation of the receiver wells is recommended. No operational changes—including choke movements, swabbing, ESP speed adjustments, or well tests—are allowed on the observation wells to preserve the integrity of the pressure data.

In addition, nearby wells whose pressure transients may interfere with the signal in the observation wells; thus nearby wells must be held at constant rate. These constraints are defined in advance based on the pulse test design review to reduce cross-interference and improve confidence in the measured interwell signal.

Pressure-Data Conditioning, Signal Decomposition, and Parameter Extraction

The interpretation workflow integrates an analytic forward model for multi-cycle pulse tests with an automated, data-driven inversion routine. The process begins with noise filtering and detrending. A time-varying detrend algorithm is applied to subtract the constant rate depletion slope that developed prior to the test, thereby eliminating the monotonic component of the reservoir response. In addition, low-band tidal and barometric oscillations are attenuated using signal filters to improve the signal-to-noise ratio in the data.

Following this, automated signal decomposition is performed at each observation well. The pressure data is processed using a correlation minimisation routine that searches for the linear combination of modelled pulse responses (generated from the forward model) that best reconstructs the observed data. This algorithm

decomposes the pressure data into two components: one is a pulse signal-related component that is highly correlated with the imposed cycles in the generator well, and the other is a residual component that captures unrelated field events, wellbore dynamics, or sensor drift.

Once decomposition is complete, parameter inversion and uncertainty quantification are carried out. The amplitudes and time lags of the pulse signal component are regressed against the analytic solution to estimate interwell transmissibility and storativity. Based on the strength of correlation with the pulse signal, connectivity is classified as good, marginal, or absent.

The final deliverables of the workflow include a summary table with estimated transmissibility, storativity including uncertainty values for every pulsing–observation well pair. In addition, spider plots are used to visualize the connectivity network, clearly highlighting potential sealed faults or stratigraphic barriers between the pulsing and observation wells.

Field A Case Study — Objectives, Challenges, and Pulse-Test Programme

Field A as example of technology application is clastic onshore oil field with multiple isolated blocks and facing a connectivity challenge between blocks. Routine surveillance and history-matching suggested that cross-fault communication was much weaker than assumed in the static model, raising concern that the existing waterflood pattern was not balancing voidage. A dedicated pulse-testing campaign was therefore designed with three main objectives: confirm the transmissibility of the main bounding faults, validate the placement of a planned infill injector, and acquire data for updating the dynamic model.

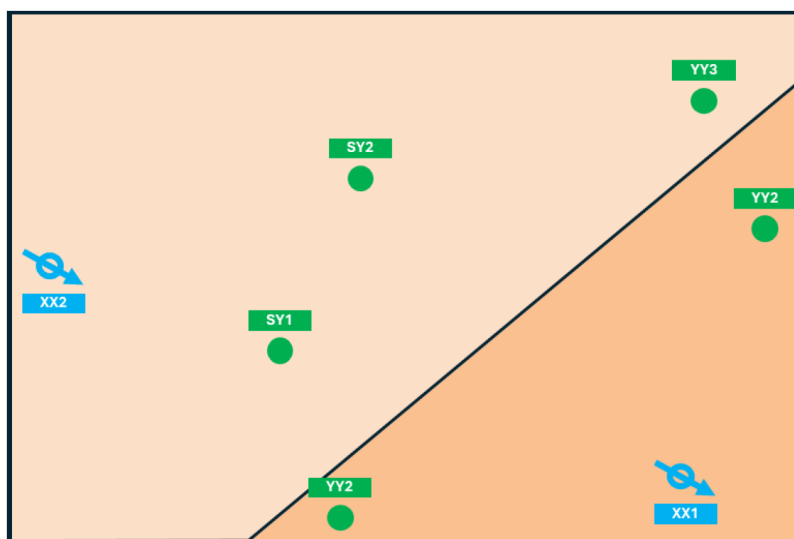


Figure 1—Field A compartment schematic illustration

The pulsing well - XX1 - is a water injector with planned injection rate of 2700 bbl/d during pulse test. Three oil producers (YY1–YY3) are selected as receivers. Because the well spacing exceeds 1.5 km and gas cap expansion, maximising the signal-to-noise ratio was critical thus number of pulses, active duration, and shut-in duration are optimized to enhance signal clarity. Well intervention, pulse test monitoring, design program, schedule and timing (Table 1, 2) was planned to minimize the risks of operations and maximize signal/noise ratio taking into account completion scheme and integrity status of the wells.

Table 1—Pulse Test Design

String Name	Role	Type/Status/Lift Method	String Mode During PCT	1 Period Time hrs. (d)	Liquid Rate (bpd)
XX1	Generator	WI/Active/-	Injecting/Shut-in (alternating)	240 (10)	2700
YY1	Receiver	OP/Active/GL	Flowing		
YY2	Receiver	OP/Active/GL	Flowing		
YY3	Receiver	OP/Active/GL	Flowing		

Table 2—Pulse Test Activity Schedule

Month	Dec						Jan	
Days	Pre-PCT	10 days	10 days	10 days	10 days	10 days	10 days	Post-PCT
XX1	TCC & set Gauges	on	off	on	off	on	off	Retrieve gauges
YY1		Keep the well active & stable						
YY2		Keep the well active & stable						
YY3		Keep the well active & stable						
	Pre-PCT	PCT Survey (60 days)						Post PCT
		on	Inject at 2700 bwpd					
		off	Shut-in the Well					

Final established and approved pulse test design program taking into account distance between wells and potentially high gas saturation, each pulse consisted of 240 h of injection followed by 240 h of shut-in, repeated three times over a 60-day survey window. Dual quartz gauges were installed below the fluid level in every receiver.

After despiking, tidal correction and removal of the slow depletion trend, the pre-processed gauge record was interpreted to detect pulse test signal strength. Final calibrated pulse test response model has high 0.86 correlation. The best-fit lag for the first cycle is X h, and the recovered pulse amplitude of Y psi. Calculated effective permeability is within 80–110 md range. Similar results were obtained for YY1, whereas YY3 has no detectable response, indicating a sealed compartment. Based on evaluated connectivity between compartments, the planned number of injectors in the northern block was optimized.

Conclusion

Reservoir characterization is critical for successful field development. Pulse-test success in complex reservoirs depends on two factors:

- generating a pressure pulse that rises above background noise and,
- applying a processing scheme that isolates that signal with confidence.

For clastic assets such as Field A with compartmentalization, gas cap and high heterogeneity – advanced reservoir surveillance programme are required to manage oil development. Analytic design tools, supported by Monte-Carlo analysis, produced a three-cycle sequence whose amplitude consistently exceeded twice the RMS tidal fluctuation observed in Field A. After despiking and trend removal, automated decomposition

recovered a pulse signature that matched the generator-well schedule with an 86 % correlation. The resulting 80–110 md effective permeability confirmed communication between two blocks while proving another block to be disconnected, supporting asset team to reduce the number of planned injectors.

By integrating rigorous design with modern signal processing, the study repositions pulse testing from a niche technique to a practical, field-wide surveillance tool that delivers reservoir understanding such as connectivity in complex assets.

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